

When M8 is just about to turn on (M8's V_{GS} is approaching V_{THN} but the drain currents of M8 and M6 are still zero), the current flowing in M7 is

$$i_{om} = \frac{\beta_B}{2}(v_{op} - V_{THN})^2 \quad (27.3)$$

and the current flowing in M5 is

$$i_{op} = \frac{\beta_A}{2}(v_{op} - V_{THN})^2 \quad (27.4)$$

Noting that the current in M7 (at the switching point) mirrors the current in M5, we can write

$$i_{op} = \frac{\beta_A}{\beta_B} \cdot i_{om} \quad (27.5)$$

If $\beta_A = \beta_B$, then switching takes place when the currents, i_{op} and i_{om} , are equal. Unequal β s cause the comparator to exhibit hysteresis. Relating these equations to Eq. (27.1) yields the switching point voltages (review Ch. 18), or

$$V_{SPH} = v_p - v_m = \frac{I_{SS}}{g_m} \cdot \frac{\frac{\beta_B}{\beta_A} - 1}{\frac{\beta_B}{\beta_A} + 1} \text{ for } \beta_B \geq \beta_A \quad (27.6)$$

and

$$V_{SPL} = -V_{SPH} \quad (27.7)$$

Consider the following example.

Example 27.1

For the circuit shown in Fig. 27.5, estimate and simulate the switching point voltages for two designs: (1) $W_5 = W_6 = W_7 = W_8 = 10$ with $L = 1$ and (2) $W_5 = W_8 = 10$ and $W_6 = W_7 = 12$ with $L = 1$.

For the first case (using the data in Table 9.1)

$$\beta_A = \beta_B = 120 \frac{\mu A}{V^2} \cdot \frac{10}{1} = 1.2 \text{ mA/V}^2$$

so that, as seen in Eq. (27.6) $V_{SPH} = V_{SPL} = 0$. In other words, the comparator does not exhibit hysteresis. Simulation results are shown in Fig. 27.6. The input v_m is set to 2.5 V, while the v_p input is swept from 2.48 to 2.52 V. Note that the amplitudes of v_{op} and v_{om} are limited.

For the second case

$$\beta_A = 120 \frac{\mu A}{V^2} \cdot \frac{10}{1} \text{ and } \beta_B = 120 \frac{\mu A}{V^2} \cdot \frac{12}{1}$$

or $\beta_B = 1.2\beta_A$. Using Eqs. (27.6) and (27.7) with $I_{SS} = 40 \mu A$ and $g_m = 150 \mu A/V$ (again, see Table 9.1), we get

$$V_{SPH} = -V_{SPL} = \frac{I_{SS}}{g_m} \cdot \frac{\frac{\beta_B}{\beta_A} - 1}{\frac{\beta_B}{\beta_A} + 1} = \frac{40}{150} \cdot \frac{1.2 - 1}{1.2 + 1} = 24 \text{ mV}$$

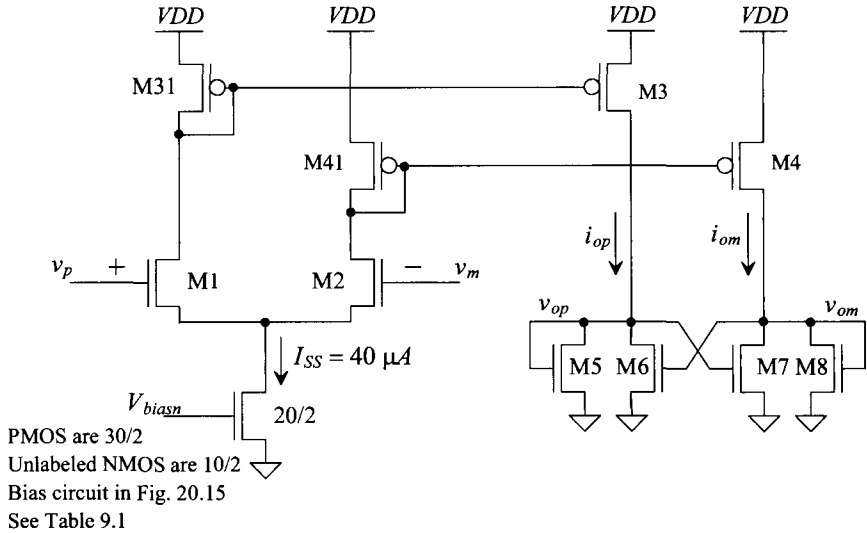
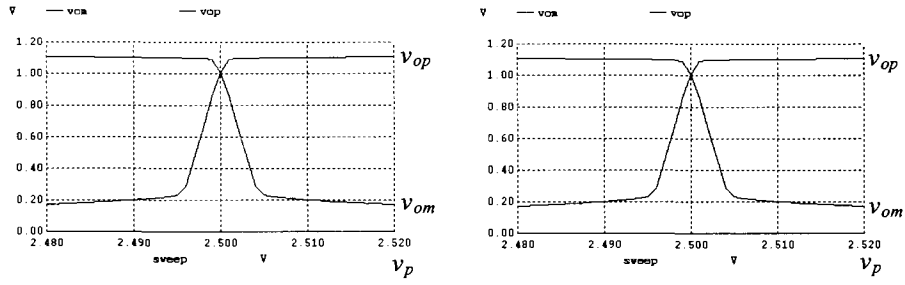


Figure 27.5 Schematic of the pre-amp and decision circuit used in Ex. 27.1.



(a) Minus input held at 2.5 V while the positive input is swept from 2.4 to 2.6 V.
 (b) Minus input held at 2.5 V while the positive input is swept from 2.52 to 2.4.

Figure 27.6 The outputs of the decision circuit in Fig. 27.5 without hysteresis.

The simulation results are shown in Fig. 27.7. Figure 27.7a shows a sweep of v_p from 2.4 to 2.6 V, with v_m held at 2.5 V. Since V_{SPH} is 24 mV, the decision circuit switches states when v_p is 24 mV above v_m (i.e., when $v_p = 2.524$ V). The case of v_p being swept from 2.6 to 2.4 V is shown in part (b) of the figure. Switching occurs in this situation when v_p is approximately 2.476 V, or 24 mV, less than 2.5 V. ■

Output Buffer

The final component in our comparator design is the output buffer or post-amplifier. The main purpose of the output buffer is to convert the output of the decision circuit into a logic signal (i.e., 0 or VDD). The output buffer should accept a differential input signal

Figure 27.9 shows a comparator design with both a PMOS and NMOS diff-amp feeding the decision circuit (for input common-mode range beyond the power supply rails). The output buffer is a PMOS diff-amp driving an inverter. While this op-amp won't be as fast as the op-amp seen in Fig. 27.8, it is a good general purpose design. The current drawn from V_{DD} will be considerably less than the design in Fig. 27.8. Note that the output currents of the PMOS diff-amp, on the input of the comparator, are added to the NMOS (current) outputs to ensure that current is always sourced to the decision circuit. We might be tempted to connect the PMOS diff-amp directly to the decision circuit. However, this would result in a larger minimum input common-mode voltage.

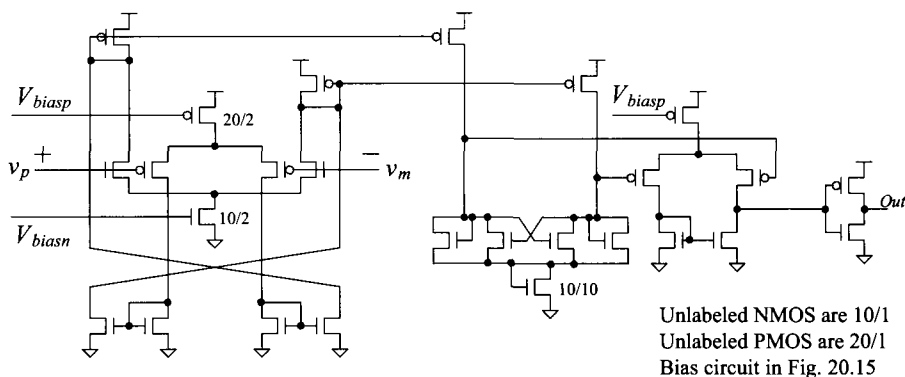


Figure 27.9 General-purpose comparator with rail-to-rail input common-mode range.

27.1.1 Characterizing the Comparator

Comparator DC Performance

Figure 27.10 shows the DC simulated performance of the comparator in Fig. 27.9. The positive comparator input is swept from 0 to V_{DD} ($= 5\text{ V}$) while the negative input is stepped for each simulation from 0 to 5 in 500 mV increments (to show wide input common-mode range). Also seen in this figure is the comparator's current draw from V_{DD} . Included in this current is the current supplied to the bias circuit.

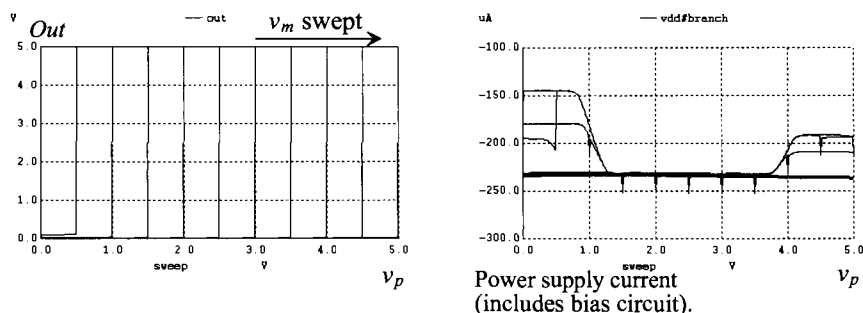


Figure 27.10 DC performance of the comparator in Fig. 27.9

If we take the derivative of these transfer curves, the gain of the comparator and thus the smallest difference that can be discriminated between v_p and v_m becomes known. We have to be careful here with the step size used in the DC simulation. If, for example, we were to use a step size of 1 mV, then the maximum gain we get from the simulation is 1,000. Figure 27.11 shows the expanded view of the comparator with the minus input held at 2.5 V while the positive input is swept from 2.499 to 2.501 V. The offset seen in the figure is a systematic offset (as discussed earlier). The gain of the comparator is, roughly, 175,000. Notice that the comparator was designed without hysteresis. However, in a practical comparator mismatch in the decision circuit results in hysteresis (and so if hysteresis isn't desired, the layout of the decision circuit is critical and may require common-centroid techniques).

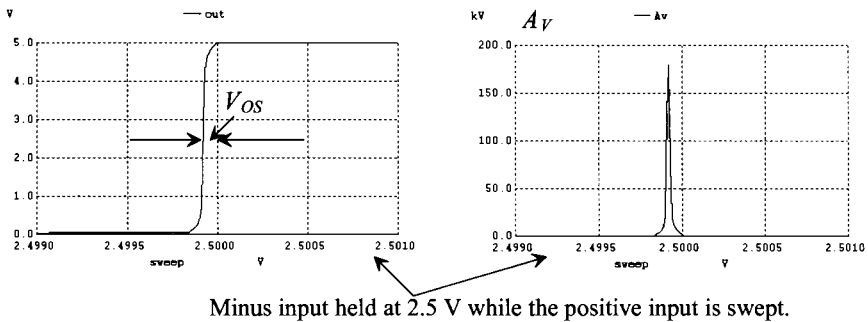


Figure 27.11 The gain of the comparator in Fig. 27.9

Transient Response

The transient response of a comparator can be significantly more difficult to characterize than the DC characteristics. Let's begin by considering $v_m = 2.5$ V DC, with the v_p input to the comparator a 10 ns wide pulse and an amplitude varying from 2.45 to 2.55 V. This is termed a narrow pulse with a 50 mV overdrive; we are driving the + input of the comparator 50 mV over the negative input. The simulation results for these inputs are shown in Fig. 27.12. If the pulse amplitude or width is reduced much beyond this, the comparator does not make a full transition. Notice how we only show simulation data after 120 ns. This was done to ensure the bias circuit had time to start-up and stabilize before changing the inputs of the comparator.

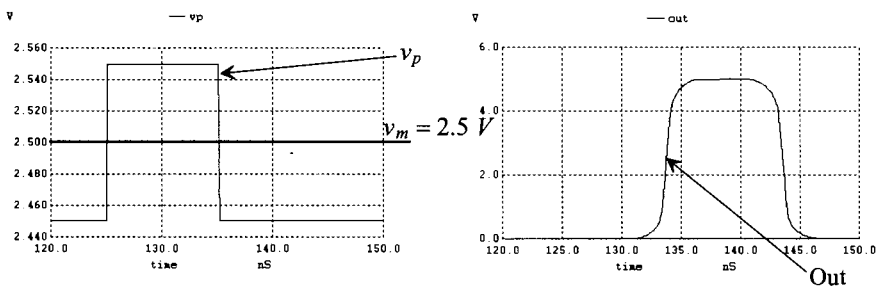


Figure 27.12 Transient response of the comparator in Fig. 27.9.

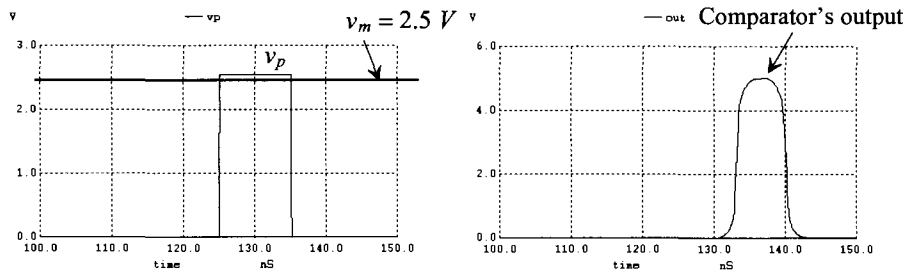


Figure 27.13 Transient response of the comparator in Fig. 27.9 with one side of the pre-amp initially shut off.

Although these results are interesting, they are not practically useful in some situations. In few cases will the comparator discriminate between signals similar to what is seen in Fig. 27.12. A better indication of comparator performance is to apply a signal that starts at a voltage where one side of the differential amplifier is cut off and to finish at a voltage slightly larger than the reference voltage (2.5 V in the above example), Fig. 27.13. In applying a 0 to 2.6 V pulse to the v_p input, the left side of the diff-amp is initially off and then on. The internal comparator nodes must be charged over a wider voltage range with a 0–2.6 V input when compared to the 2.45–2.55 V example of Fig. 27.12. In order to keep one side of the pre-amp from turning off, the circuit of Fig. 27.14 can be used. The added MOSFET ensures that the voltage between the drains of M31 and M41 don't deviate too much. This configuration is sometimes referred to as a clamped input stage. Note that the size of the added MOSFET should be as small as possible to avoid unneeded loading in the diff-amp (which slows down the comparator's performance).

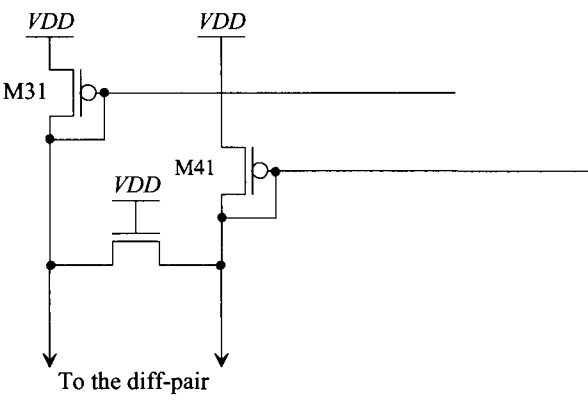


Figure 27.14 Adding a balancing resistor to the input stage of a comparator. See Fig. 27.5

Propagation Delay

Ideally, the propagation delay (the time difference between the input, v_p , crossing the reference voltage, v_m , and the output changing logic states) is zero. For the comparator of Fig. 27.9 with simulation results shown in Fig. 27.12, the delay of the comparator is approximately 9 ns (or less with an overdrive). This should be compared with the delay of an op-amp used as a comparator, which may be several hundred nanoseconds (because of the compensation capacitor). Another interesting fact about comparator design is that the delay of a comparator can be reduced by cascading gain stages. In other words, the delay of a single high-gain stage is in general longer than the delay of several low-gain stages. Improvements in comparator sensitivity and speed can be directly related to improvements in the preamplifier. A derivation (similar to that given in Ch. 11 for minimum delay through an inverter string driving a load capacitance, C_L) of the number of stages, N , needed in a preamplifier to reach minimum delay for a given load capacitance and input capacitance results in

$$N = \ln \frac{C_L}{C_{in}} \quad (27.8)$$

In the derivation of this equation, it was assumed that the driving resistance of any stage is $1/g_{mn}$ (the transconductance of the n^{th} differential amplifier stage). In practice, Eq. (27.8) is of little use because one side of the differential amplifier can be off. For this reason (and others such as size and power draw), comparators with more than two or three pre-amp stages are not common.

Minimum Input Slew Rate

The last characteristic we discuss is the minimum input slew rate of a comparator. If the input signals to the comparator vary at a slow rate (e.g., a sine wave generated from the AC line), the output of the comparator may very well oscillate resulting in an output with a metastable state. If a comparator is to be used with slowly varying signals, or in a noisy environment, the decision circuit should have hysteresis. The minimum input slew rate is difficult to simulate with SPICE because of the slow and fast varying signals present at the same instant of time in the circuit. This same situation (metastability) can also occur if the input overdrive is small. The result, for this case, however, is an increase in the delay time of the comparator.

27.1.2 Clocked Comparators

We developed “sense amplifiers” in Sec. 16.2.1. The sense amplifier can be used as a clocked comparator, that is a comparator whose outputs change on the rising (or falling) edge of a clock. Consider the clocked comparator in Fig. 27.15. This circuit is directly taken from Figs. 16.32 and 16.35. It works very well for signal differences that are tens of millivolts. When *clock* is low, the inputs to the NAND SR latch are pulled high. The outputs of the comparator don’t change. When *clock* goes high, the two inputs are compared causing the output of the circuit to register which one is higher. As discussed in detail back in Ch. 16, the minimum input voltages for this comparator are above the threshold voltage of the NMOS devices. Further if the input signals get too large, MB1 and MB2 move deep into the triode region. The small differences in their channel resistances adversely affect the quality of the comparator’s decisions. Let’s modify this topology for better sensitivity, wider input signal swing, and better immunity to kickback.

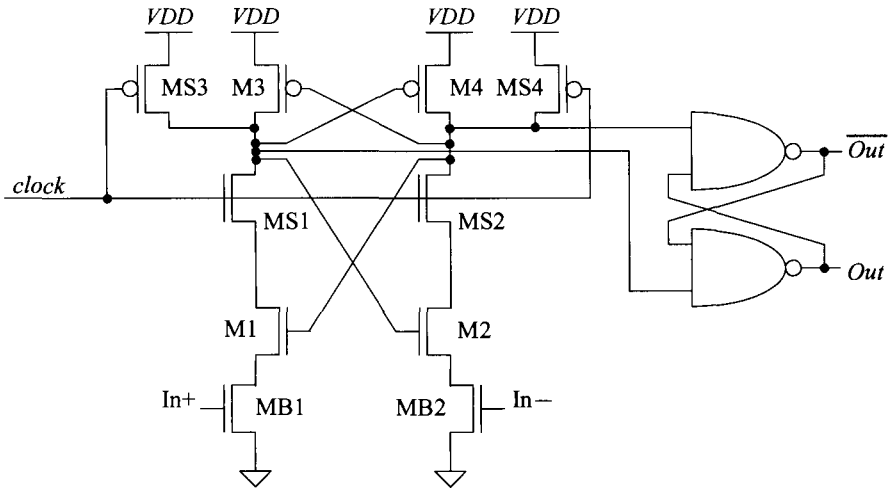


Figure 27.15 A clocked comparator based on Figs. 16.32 and 16.35.

Our modification is seen in Fig. 27.16. Instead of biasing the PMOS and NMOS diff-amps with a bias circuit, here, to be different, we simply use long-length MOSFETs. The bias currents aren't too critical in this application (like they are in an op-amp design). We can still use a bias circuit if controlling the bias current is, for some reason, important. We've removed the triode-operating MOSFETs MB1 and MB2 from the basic cross-coupled latch section. Now we are steering currents from one side of the latch to the other. The differences in the currents causes the latch to switch dependent on the input signals. Note that if either of the diff-amps isn't present in the comparator or if one is off because the comparator's input common-mode voltage is too high or too low, the comparator still works as desired. Only one diff-amp is needed to create the imbalance. Also note, again, that the SR latch is used to make the outputs of the circuit change on the rising edge of the clock signal. If the outputs of the comparator can go high, for a particular application, when the clock signal is low then the NAND gates can be removed.

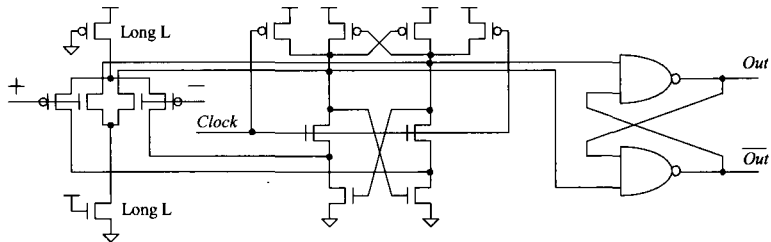


Figure 27.16 Wide-swing clocked comparator. Outputs change on the rising edge of the clock signal.

27.1.3 Input Buffers Revisited

When we discussed input buffers back in Ch. 18 we used a self-biased design for the highest speed (see Fig. 18.17 and the associated discussions). As discussed in Ch. 18, it's desirable to have symmetrical propagation delays independent of input slew-rate, amplitudes, or direction (high-to-low or low-to-high). Towards the more ideal input buffer, consider using two NMOS self-biased buffers in parallel, as seen in Fig. 27.17. Here, to maintain good symmetry, we've split the current sources in half and used one side to generate a common-mode feedback signal to balance the outputs. To reduce the power dissipation in the buffer, the triode-operating MOSFETs can have their lengths increased. Further, for rail-to-rail input common-mode range, we can place the PMOS version of this buffer in parallel with the NMOS version seen in Fig. 27.17, as we did in Fig. 18.23. This input buffer can be very useful for low-skew, high-speed design.

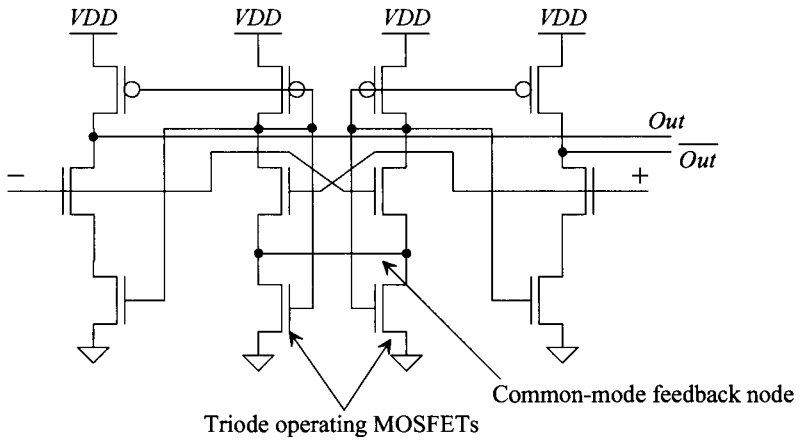


Figure 27.17 A fully-differential input buffer based on the topology discussed in Sec. 18.17.

27.2 Adaptive Biasing

Adaptive biasing can reduce power dissipation in an amplifier while at the same time increasing output current drive capability. Figure 27.18 can be used to help illustrate the idea. When v_{i1} and v_{i2} are equal, the current sources I_{SS1} and I_{SS2} are zero (an open). The diff-amp DC tail current is simply I_{SS} , the same as an ordinarily biased diff-amp. If v_{i1} becomes larger than v_{i2} , the current source I_{SS1} increases above zero, effectively increasing the diff-amp DC bias current. Similarly, if v_{i2} becomes larger than v_{i1} , the current source I_{SS2} increases above zero. The diff-amp output current is normally limited to I_{SS} when one side of the diff-amp shuts off. However, now that the maximum output current is limited to either $I_{SS} + I_{SS1}$ or $I_{SS} + I_{SS2}$. Power dissipation can be reduced using an adaptive bias, and slew-rate problems can be eliminated.

If $v_{I1} = v_{I2}$ then $I_{SS1} = I_{SS2} = 0$.

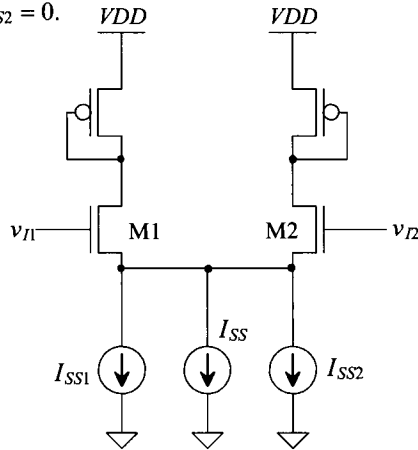


Figure 27.18 Adaptively biased diff-amp.

The current diff-amp of Fig. 27.19 can be used to implement the current source I_{SS1} or I_{SS2} . If the currents I_1 and I_2 are equal, then zero current flows in M3 and M4. Also, if I_2 is greater than I_1 , zero current flows in M3 and M4. If I_1 is larger than I_2 , the difference between these two currents ($I_1 - I_2$) flows in M3. Since M4 is K times wider than M3, a current of $K(I_1 - I_2)$ flows in M4 (normally $K < 1$). Two of these diff-amps are needed to implement the adaptive biasing of the diff-amp of Fig. 27.18.

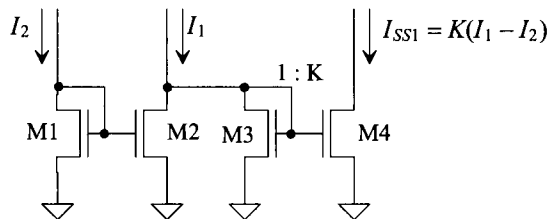


Figure 27.19 Current diff-amp used in adaptive biasing.

Figure 27.20 shows the implementation of adaptive biasing into the diff-amp of Fig. 27.18. P-channel MOSFETs are added adjacent to M3 and M4 to mirror the currents through M1 and M2 (I_1 and I_2). The maximum total current available through M1 occurs when M2 is off. Positive feedback exists through the loop M1, M3, M5–M7. Initially, when M2 shuts off, the current in M1 and M3 is I_{SS} . This is mirrored in M5 and M6, and thus I_{SS1} becomes $K \cdot I_{SS}$. At this particular instance in time, the tail current, which flows through M1, is now $I_{SS} + K \cdot I_{SS}$. However, provided the MOSFETs M1, M3–M7 remain in

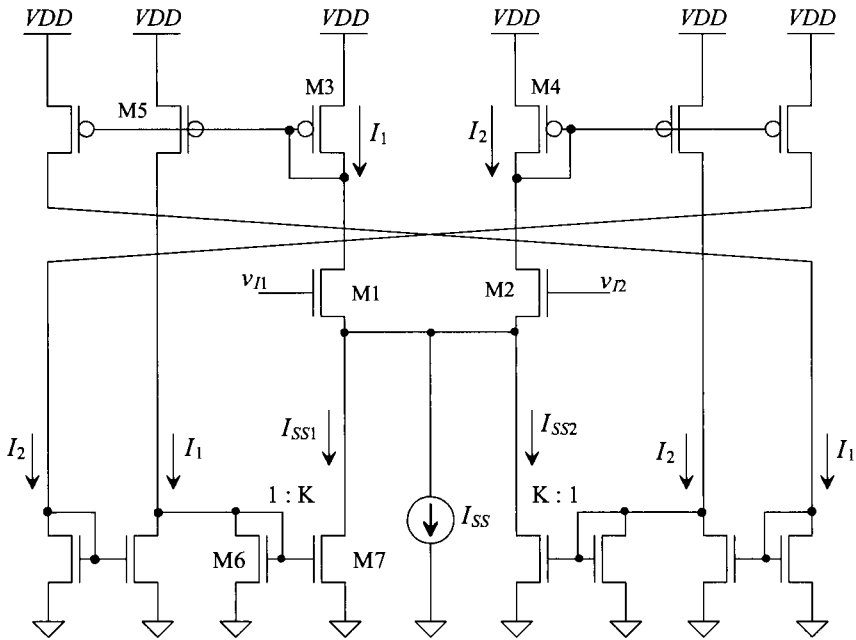


Figure 27.20 Adaptively biased diff-amp.

saturation, this current circles back around the positive feedback loop and increases by K . This continues, resulting in a final or total tail current of

$$I_{tot} = I_{SS} \cdot (1 + K + K^2 + K^3 + \dots) \quad (27.9)$$

If $K < 1$, this geometric series can be written as

$$I_{tot} = \frac{I_{ss}}{1 - K} \quad (27.10)$$

Setting $K = 0$ (MOSFET M7 doesn't exist) results in no adaptive biasing and a total tail current of I_{SS} . Setting $K = 1/2$ (M7 half the size of M6) results in a total available tail current of $2 \cdot I_{SS}$. Because the tail current limits the slew rate, when the diff-amp is driving a capacitive load, making K equal to one eliminates slew-rate limitations while at the same time not increasing static power dissipation. In practice, shutting off one side of the diff-pair, M1/M2, is difficult since the adaptive biasing has the effect of lowering the source potentials of the diff-pair, keeping both MOSFETs on. Adaptive biasing can be used in a comparator where M1 or M2 can shut off. However, the static power dissipation will be large; therefore, there is no benefit over the comparators discussed earlier in the chapter. When applying adaptive biasing to an OTA design, the value of K should be unity or less. [Using a K of 1 or 2 can still result in a finite I_{tot} since the MOSFETs have a finite output resistance and the MOSFETs in the diff-pair will not shut off, as was assumed in the derivation of Eq. (27.10)].